

12 (**short**) Consider the following three C++ code segments.

```
// Segment 1
for (i=0; i<n; i++)
    for (j=0; j<n; j++)
        m2[j][i] = m1[i][j];

// Segment 2
for (i=0; i<n; i++)
    v3[i] = v2[i] + v1[i];

// Segment 3
for (i=0; i<n; i++)
    for (j=0; j<n; j++) {
        m3[i][j] = 0;
        for (k=0; k<n; k++)
            m3[i][j] += m2[i][k] * m1[k][j];
    }
```

(a) For each code segment, state

(i) its purpose and

(ii) its algorithmic complexity.

[7]

(b) All three segments are executed sequentially, taking a total time of 1 s when n is 1000. Estimate how long, in total, they would take to execute when n is 2000.

[3]

END OF PAPER

Answers

1 $0, \pm 3\sqrt{3}i$

2 Intersections at $-2.07, -4.49$

3 (a) $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

4 (a) $(3, 1, 2)$ (c) α (d) $\frac{6}{\sqrt{5}}$

5 (a) $\frac{1}{4}(1 - \cos 2t)$ (b) $-\frac{1}{2}\cos 2t$ (c) $\frac{1}{40}(1 - 19\cos 2t)$

6 te^{-t}

7 (b) $\frac{2}{3} - \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \cos n\pi x$ (c) $\sum_{n=1}^{\infty} 4 \cos n\pi x$

8 (a) $1 - 2p + 2p^2$

9 $(1, 1)$, saddle

10 (b) $2e^{-t} - e^{-t} \cos t$ (c) $\frac{1}{2}(\sin t - t \cos t)$ (d) $\frac{e^{-t}}{2}(\sin t - t \cos t)$

12 (b) $8s$